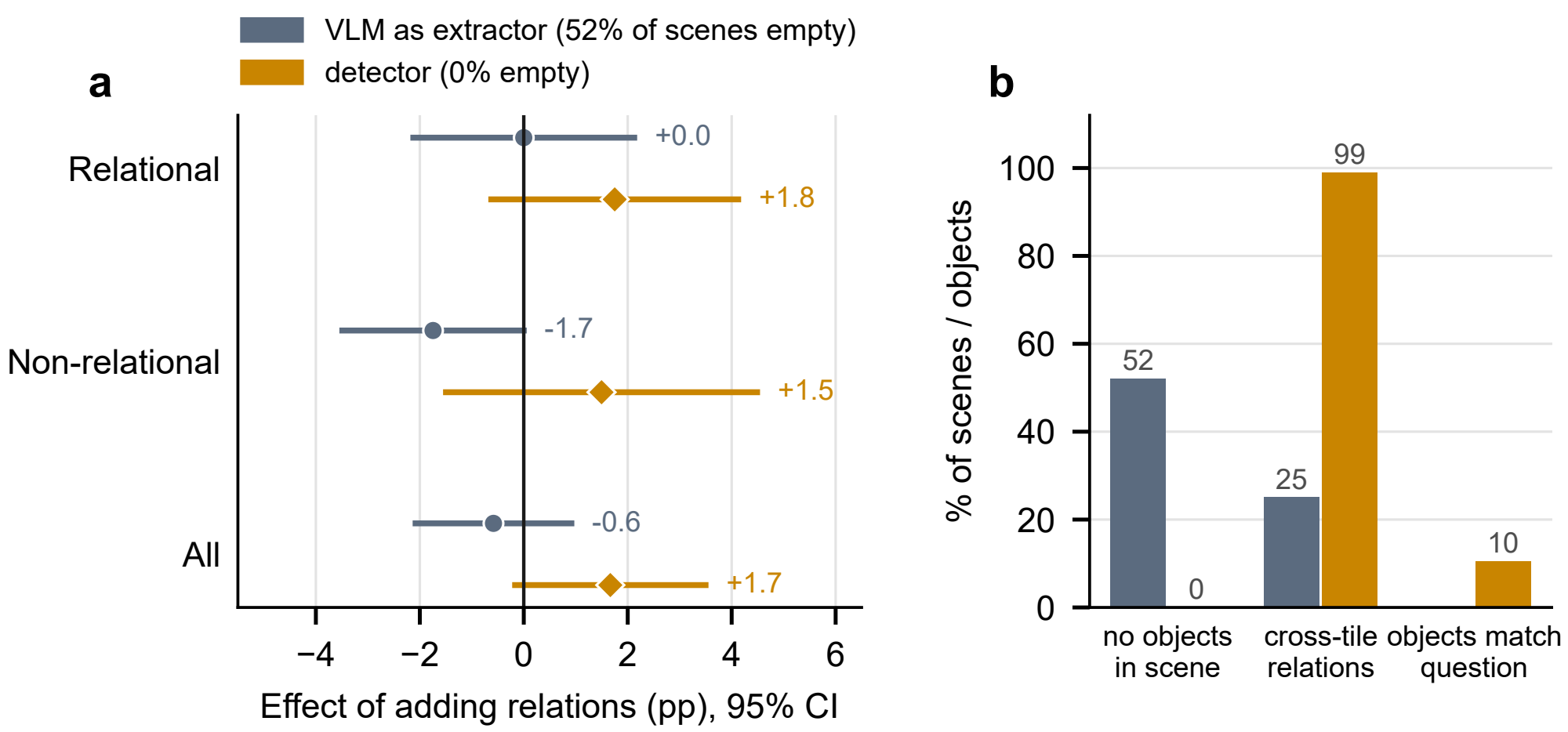
a
VLM as extractor (52% of scenes empty)
detector (0% empty)
Relational
Non-relational
All
+0.0
+1.8
-1.7
+1.5
-0.6
+1.7
−4
−2
0
2
4
6
Effect of adding relations (pp), 95% CI
b
% of scenes / objects
100
80
60
40
20
0
52
0
25
99
10
no objects in scene
cross-tile relations
objects match question